

Chat Link: https://parley.mit.edu/share/3XuxwWZJJY9P_NZ6LaDPE

```
global_data = {
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "mass",
  "attribute_4": "connection_reliability",
  "attribute_5": "cross_sectional_area",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "negative",
  "direction_attribute_4": "positive",
  "direction_attribute_5": "negative",

  "attribute_1_min": 75,
  "attribute_1_max": 900,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 8,
  "attribute_3_max": 60,
  "attribute_4_min": 0,
  "attribute_4_max": 10,
  "attribute_5_min": 5,
  "attribute_5_max": 90,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.500,    # fixed global Cu price
  "aluminum_price_per_kg": 2.500, # fixed global Al price
}
```

```
regional_data = [
  {
    "region": "United States",
    "copper_product_market_share": 0.90,
    "aluminum_product_market_share": 0.10,
    "weight_attribute_1": 0.30,
    "weight_attribute_2": 0.20,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.20,
    "weight_attribute_5": 0.10
  },
]
```

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{
  "region": "China",
  "copper_product_market_share": 0.70,
  "aluminum_product_market_share": 0.30,
  "weight_attribute_1": 0.40,
  "weight_attribute_2": 0.10,
  "weight_attribute_3": 0.20,
  "weight_attribute_4": 0.15,
  "weight_attribute_5": 0.15
},
{
  "region": "India",
  "copper_product_market_share": 0.95,
  "aluminum_product_market_share": 0.05,
  "weight_attribute_1": 0.45,
  "weight_attribute_2": 0.15,
  "weight_attribute_3": 0.10,
  "weight_attribute_4": 0.20,
  "weight_attribute_5": 0.10
},
{
  "region": "Europe",
  "copper_product_market_share": 0.80,
  "aluminum_product_market_share": 0.20,
  "weight_attribute_1": 0.20,
  "weight_attribute_2": 0.15,
  "weight_attribute_3": 0.30,
  "weight_attribute_4": 0.20,
  "weight_attribute_5": 0.15
}
]

```

```

product_data = [
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "ICE Wire Harness",
    "copper_kg": 20,
    "aluminum_kg": 0,
    "total_mass_kg": 28,
    "non_material_cost_per_unit": 280,
    "attribute_2_value": 1,
    "attribute_3_value": 28,
    "attribute_4_value": 8.5,

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"attribute_5_value": 30
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "ICE Wire Harness",
  "copper_kg": 0.5,
  "aluminum_kg": 12,
  "total_mass_kg": 21,
  "non_material_cost_per_unit": 340,
  "attribute_2_value": 0,
  "attribute_3_value": 21,
  "attribute_4_value": 6.5,
  "attribute_5_value": 48
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "ICE Wire Harness",
  "copper_kg": 20,
  "aluminum_kg": 0,
  "total_mass_kg": 28,
  "non_material_cost_per_unit": 150,
  "attribute_2_value": 1,
  "attribute_3_value": 28,
  "attribute_4_value": 8.0,
  "attribute_5_value": 30
},
{
  "region": "China",
  "dominant material": "al",
  "product": "ICE Wire Harness",
  "copper_kg": 0.5,
  "aluminum_kg": 12,
  "total_mass_kg": 21,
  "non_material_cost_per_unit": 180,
  "attribute_2_value": 0,
  "attribute_3_value": 21,
  "attribute_4_value": 6.0,
  "attribute_5_value": 48
},
{
  "region": "India",
  "dominant material": "cu",
```

```
"product": "ICE Wire Harness",
"copper_kg": 20,
"aluminum_kg": 0,
"total_mass_kg": 28,
"non_material_cost_per_unit": 120,
"attribute_2_value": 1,
"attribute_3_value": 28,
"attribute_4_value": 8.0,
"attribute_5_value": 30
},
{
  "region": "India",
  "dominant material": "al",
  "product": "ICE Wire Harness",
  "copper_kg": 0.5,
  "aluminum_kg": 12,
  "total_mass_kg": 21,
  "non_material_cost_per_unit": 145,
  "attribute_2_value": 0,
  "attribute_3_value": 21,
  "attribute_4_value": 5.5,
  "attribute_5_value": 48
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "ICE Wire Harness",
  "copper_kg": 20,
  "aluminum_kg": 0,
  "total_mass_kg": 28,
  "non_material_cost_per_unit": 380,
  "attribute_2_value": 1,
  "attribute_3_value": 28,
  "attribute_4_value": 9.0,
  "attribute_5_value": 30
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "ICE Wire Harness",
  "copper_kg": 0.5,
  "aluminum_kg": 12,
  "total_mass_kg": 21,
  "non_material_cost_per_unit": 460,
```

```
"attribute_2_value": 0,  
"attribute_3_value": 21,  
"attribute_4_value": 7.0,  
"attribute_5_value": 48  
}  
]
```